PURE 1. # Quadratics

1 Factorising quadratics

$y = ax^2 + bx + c$

(a, b and c are numbers)

when $a = 1$

$ax^2 + bx + c$

$x^2 + 7x + 12 = 0$

$(x + \quad)(x + \quad) = 0$

2 numbers that add / subtract = b (7)
multiply = c (12)

$(x + 3)(x + 4) = 0$

$x = -3$ $\qquad$ $x = -4$

when $a \neq 1$, you need more brain juice 

$2x^2 + 7x + 6 = 0$

$(2x \quad)(x \quad) = 0$

$(2x + 3)(x + 2)$

$x = -\frac{3}{2}$ $\qquad$ $x = -2$

to get +6

2×3 $\qquad$ OR $\qquad$ -2×-3

3×2 $\qquad\qquad$ -3×-2

2. Completing the Square

↳ for sketching graphs ; solving quadratic equations , circle equations

$ax^2 + bx + c \longrightarrow \underbrace{a\left(x + \frac{b}{2a}\right)^2}$ $+ d$ ← the adjusting number

bx: divide by 2

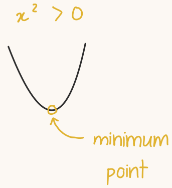
multiply out
↳ adjust to equal original quadratic

$\underbrace{x^2 + 12x} + 16$

$\underbrace{(x+6)^2}$

$x^2 + 12x + 36 + d \Rightarrow d = -20 \longrightarrow = (x+6)^2 - 20$

↳ Sketch the Graph using the Completed Square



* quadratics have **1** turning point

y-intercept

$y = x^2 - 14 + 36 \quad = \quad (x-7)^2 - 13$

turning point info
(7, -13)

factorise OR complete the square

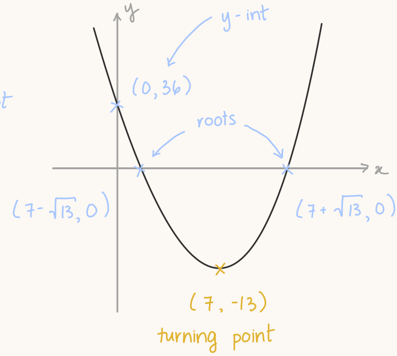
$x - 7 = 0$

$x = 7$ ← x-coordinate

y-coordinate

$x = 7 + \sqrt{13}$; $x = 7 - \sqrt{13}$

roots / x-intercept



3. The quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

discriminant
$b^2 - 4ac$

↳ how many roots?

> 0 ⇒ 2 real roots
= 0 ⇒ 1 real root
< 0 ⇒ no (real) roots

4. Quadratic Inequalities

- **Inequalities**:
  - do the same thing on both sides
  - if you multiply / divide by a negative number
    ↳ FLIP the inequality sign

$3x + 4 > 2x - 1$

$x > -5$

$4x \geq 8x + 12$

$-4x \geq 12$

$x \leq -3$

$-5 < x \leq -3$

$>$ or $<$ = open circle ○

$\geq$ or $\leq$ = closed circle ●

* quadratic inequalities: solutions can be

2 sets of values

$3x^2 - 22 > 5$

$3x^2 > 27$

$x^2 > 9$

$x > 3$; $x < -3$

range enclosed by 2 values

$4x^2 - 14 \leq 2$

$4x^2 \leq 16$

$x^2 \leq 4$

$x \leq 2$; $x \geq -2$

↳ $-2 \leq x \leq 2$

* sketch a graph to help solve quadratic inequalities

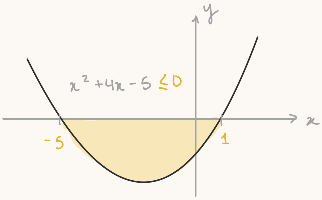
$-x^2 - 2x + 4 \geq 2x - 1$

$x^2 + 4x - 5 \leq 0$ ⟶ ∪ - shaped

$(x+5)(x-1) = 0$

$x = -5$; $x = 1$

roots



Graphing inequalities in 4 steps

1. Change the $<$ $>$ $\leq$ $\geq$ to a =
2. Draw a graph
3. Choose the correct side
   by plugging in a coordinate
   (e.g.: origin (0,0))
   that satisfies the inequality
4. Shade or label the correct region

$3x < 6 + y$

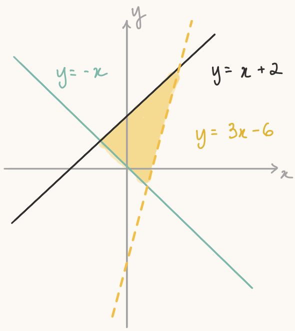
↳ $y = 3x - 6$

$-2 + y \leq x$

↳ $y = x + 2$

$y \geq -x$

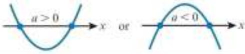
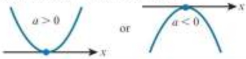
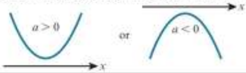
↳ $y = -x$



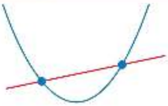
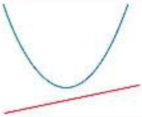
5. Number of roots of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

discriminant: $b^2 - 4ac$

b^2-4ac	Nature of roots of $ax^2+bx+c=0$	Shape of curve $y=ax^2+bx+c$
>0	two distinct real roots	The curve cuts the x-axis at two distinct points. 
$=0$	two equal real roots (or 1 repeated real root)	The curve touches the x-axis at one point. 
<0	no real roots	The curve is entirely above or entirely below the x-axis. 

6. Intersection of line and quadratic curve

Situation 1	Situation 2	Situation 3
		
two points of intersection	one point of intersection	no points of intersection
The line cuts the curve at two distinct points.	The line touches the curve at one point. This means that the line is a **tangent** to the curve.	The line does not intersect the curve.

b^2-4ac	Nature of roots	Line and curve
>0	two distinct real roots	two distinct points of intersection
$=0$	two equal real roots (repeated roots)	one point of intersection (line is a tangent)
<0	no real roots	no points of intersection